WORKSHEET

Chapter 8: Differential equations and exponentials

1 Find the general solutions to each of the following differential equations.

a $\frac{dy}{dx} = 7y$

b $\frac{dy}{dx} = y(x - 3)$

c $\frac{dy}{dx} = x + xy$

d $\frac{dy}{dx} = (3y - 1)(2x + 5)$

e $\frac{dy}{dx} + 2y = 3$

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f $\frac{dy}{dx} = (2y + 3)(4x - 1)$

2 Find the solution to each of the following differential equations.

a $\frac{dy}{dx} = xy$, if $y(2) = 4$

b $\frac{dy}{dx} = x + xy^2$, if $y(0) = 1$

c $\frac{dy}{dx} - \frac{1}{2}y = \frac{3}{2}$, if $y(0) = 2$

d $\frac{dy}{dx} - 2xy = x$, if $y(0) = 0$

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3 The half-life of a particular radioactive substance is known to be 1690 years. The substance decays at a rate proportional to the quantity present. Use a differential equation to find how much will remain of 1 gram of the substance after 1000 years.

4 Bacteria grow by division of cells, so the growth of a bacterial colony is proportional to the number present. A particular colony of bacteria grows exponentially with a growth rate of 0.02, where time is measured in hours. If there are 1500 bacteria initially, how many would be present after 1 hour?

5 In a town of 5000 people, no one has immunity to a new strain of the flu. If you come into contact with it, then you have a certain probability of becoming infected. If there are N people infected with the flu, then the number of contacts between infected and uninfected people in a day is given by $N(5000 - N)$.

We can write that the rate of change of the number of infected people is given by:

$$\frac{dN}{dt} = 0.0001 \times N(5000 - N)$$

If one person has the flu to begin with, solve the differential equation to find how many people have been infected after t days. Use your solution to find the number of infected people after 14 days.

6 An amount of $5000 is invested in an account paying 8% compounded monthly. How much money will be in the account after 5 years?

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Answers

1 **a** $y = Ae^{7x}$

b $y = Ae^{\frac{1}{2}x^2 - 3x}$

c $y = Ae^{\frac{1}{2}x^2} - 1$

d $y = \frac{1}{3} + Ae^{3x^2 + 15x}$

e $y = \frac{3}{2} + Ae^{-2x}$

f $Ae^{4x^2 - 2x} - \frac{3}{2}$

2 **a** $y = 4e^{\frac{1}{2}x^2 - 2}$

b $y = \tan\left(\frac{1}{2}x^2 + \frac{\pi}{4}\right)$

c $y = \frac{1}{2}\left(3 - e^{\frac{1}{2}x}\right)$

d $y = \frac{1}{2}\left(e^{x^2} - 1\right)$

3 About 0.6636 grams.

4 About 1530.

5 $N = \frac{1.0002e^{0.5t}}{0.0002e^{0.5t} + 1}$; 900

6 $7449.23

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